

STAT 516

Conditional Distributions and Conditional Expectations

Prof. Michael Levine

April 10, 2024

Motivation

- ▶ Let (X, Y) have the joint pmf $p(x, y)$
- ▶ The conditional distribution of X given $Y = y$ is defined to be

$$p(x|y) = P(X = x|Y = y) = \frac{p(x, y)}{p_Y(y)}$$

- ▶ The conditional expectation of X given $Y = y$ is defined to be

$$E(X|Y = y) = \sum_x xp(x|y) = \frac{\sum_x xp(x, y)}{p_Y(y)} = \frac{\sum_x xp(x, y)}{\sum_x p(x, y)}$$

Basic properties of conditional expectation

- ▶ Often, $E(X|y)$ is used a shorthand for $E(X|Y = y)$
- ▶ Let X and Y be random variables defined on a common sample space Ω . Then,

1.

$$E(g(Y)|Y = y) = g(y)$$

for any function g

2.

$$E(Xg(Y)|Y = y) = g(y)E(X|Y = y)$$

for any y and any function g

Independence of two discrete random variables

- ▶ The following are several equivalent definitions that can be used
 1. $p(x|y) = p_X(x)$ for any x, y such that $p_Y(y) > 0$
 2. $p(y|x) = p_Y(y)$ for any x, y such that $p_X(x) > 0$
 3. $p(x, y) = p_X(x)p_Y(y)$ for any x, y
 4. $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$ for any x, y
- ▶ The version (3) is usually the most convenient
- ▶ Suppose X and Y are independent random variables. Then, for any appropriate function $g(X)$ and any y ,

$$E[g(X)|Y = y] = E[g(X)]$$

Example I

- ▶ Consider the joint distribution of (X, Y) where X is the larger and Y is the smaller of the two dice rolls
- ▶ By definition,

$$P(Y = 1|X = 2) = \frac{P(Y = 1, X = 2)}{P(X = 2)} = \frac{1/18}{1/18 + 1/36} = \frac{2}{3}$$

- ▶ In the same way,

$$P(Y = 2|X = 2) = \frac{P(Y = 2, X = 2)}{P(X = 2)} = \frac{1/36}{1/18 + 1/36} = \frac{1}{3}$$

- ▶ In an automatic way, $P(Y = 1|X = 1) = 1$

Example II

- ▶ Suppose X and Y are the binary variables, each taking only values 0 and 1
- ▶ The joint distribution is

Y		
X	0	1
0	s	t
1	u	v

Example II

- ▶ By definition,

$$E(X|Y = 0) = \frac{0 \times p(0, 0) + 1 \times p(1, 0)}{p(0, 0) + p(1, 0)} = \frac{u}{s + u}$$



$$E(X|Y = 1) = \frac{0 \times p(0, 1) + 1 \times p(1, 1)}{p(0, 1) + p(1, 1)} = \frac{v}{t + v}$$

- ▶ Therefore,

$$E(X|Y = 1) - E(X|Y = 0) = \frac{v}{t + v} - \frac{u}{s + u} = \frac{vs - ut}{(t + v)(s + u)}$$

- ▶ The above implies the single formula:

$$E(X|Y = y) = \frac{u}{s + u} + \frac{vs - ut}{(t + v)(s + u)}y$$

- ▶ Whenever X and Y are both binary variables, $E(X|Y = y)$ is linear in y

Conditional variance

- ▶ Let (X, Y) have the joint pmf $p(x, y)$. Let $\mu_X(y) = E(X|Y = y)$. The conditional variance of X given $Y = y$ is

$$\text{Var}(X|Y = y) = E[(X - \mu_X)^2|Y = y] = \sum_x (x - \mu_X(y))^2 p(x|y)$$

- ▶ The usual shorthand for $\text{Var}(X|Y = y)$ is $\text{Var}(X|y)$.

Example

- ▶ Again, let X be the max and Y be the min of two rolls of a die.
- ▶ For example, if $y = 3$, then one can compute $\mu_X(y) = E(X|Y = y) = E(X|Y = 3) = 4.71$
- ▶ Therefore,

$$\begin{aligned} \text{Var}(X|y) &= \sum_x (x - 4.71)^2 p(x|3) \\ &= \frac{(3 - 4.71)^2 \times \frac{1}{36} + (4 - 4.71)^2 \times \frac{1}{18}}{\frac{1}{36} + \frac{1}{18} + \frac{1}{18} + \frac{1}{18}} \\ &\quad + \frac{(5 - 4.71)^2 \times \frac{1}{18} + (6 - 4.71)^2 \times \frac{1}{18}}{\frac{1}{36} + \frac{1}{18} + \frac{1}{18} + \frac{1}{18}} \\ &= 1.06 \end{aligned}$$

Poisson conditional distribution

- ▶ Let X and Y be the independent Poisson random variables with means λ and μ .
- ▶ The conditional distribution of X given $X + Y = t$ is $\text{Bin}(t, p)$ where $p = \frac{\lambda}{\lambda + \mu}$
- ▶ This result is very important in many applications

Proof

- ▶ Clearly, $P(X = x | X + Y = t) = 0$ if $x > t$
- ▶ For any $x \leq t$,

$$\begin{aligned} P(X = x | X + Y = t) &= \frac{P(X = x, X + Y = t)}{P(X + Y = t)} \\ &= \frac{P(X = x, Y = t - x)}{P(X + Y = t)} \\ &= \frac{e^{-\lambda} \lambda^x}{x!} \frac{e^{-\mu} \mu^{t-x}}{(t-x)!} \frac{t!}{e^{-(\lambda+\mu)} (\lambda + \mu)!} \\ &= \frac{t!}{x!(t-x)!} \frac{\lambda^x \mu^{t-x}}{(\lambda + \mu)!} \\ &= \binom{t}{x} \left(\frac{\lambda}{\lambda + \mu} \right)^x \left(\frac{\mu}{\lambda + \mu} \right)^{t-x} \end{aligned}$$

- ▶ In the above, we used the fact that $X + Y \sim \text{Poi}(\lambda + \mu)$